

MORNING REPORT

A 22-Year-Old Man Coughs Blood

Vicente Morales Oyarvide, M.D., M.P.H., and Mridula Nadamuni, M.D., from the University of Texas Southwestern Medical Center Internal Medicine Residency Program

Abstract

Pierre O. Ankomah, M.D., Ph.D.,
Editor

Morning Report is a time-honored tradition where physicians-in-training present cases to their colleagues and clinical experts to collaboratively examine an interesting patient presentation. The Morning Report section seeks to carry on this tradition by presenting a patient's chief concern and story, inviting the reader to develop a differential diagnosis and discover the diagnosis alongside the authors of the case.

A 22-year-old man sought evaluation after coughing up foul-smelling, brown sputum mixed with bright-red blood. Symptoms were preceded by chest pain in the left parasternal region that was worse on deep inspiration. Using questions, physical examination, and testing, an illness script for the presentation emerges. The differential is refined until a final diagnosis is made.

Reason for presentation: coughing up blood

Part 1: The Patient's Story

History of Present Illness: A 22-year-old man presented to the emergency department after coughing up blood-tinged sputum. Two weeks before, he noted dental pain and sensitivity and received a diagnosis of an infected wisdom tooth. A dentist removed the tooth. The patient recalled swallowing large amounts of foul-tasting saliva during the procedure. He subsequently received treatment with a 1-week course of antibiotics. He did not recall the name of the antibiotic prescribed but completed the entire regimen. He recovered rapidly after the tooth removal and did not have any additional dental pain. Three days before presentation, chest pain developed in the left parasternal region that was sharp, episodic, and worse on deep inspiration but did not radiate. On the day of presentation, the patient started coughing, expectorating foul-smelling brown sputum. Later in the day, he noticed bright-red blood mixed with the sputum produced during one of the coughing spells, and this prompted him to seek medical care. He did not have any fevers, chills, night sweats, weight loss, or dyspnea. Additional information about this patient is shown in [Box 1](#).

Box 1: Medical and Surgical History, Medications, Allergies, and Social History

Medical and Surgical History:

Recent oral surgery

Medications:

None

Allergies:

No known drug allergies.

Social History:

The patient was born in rural Mexico and moved to the southwestern United States a few years before presentation. He lives with family members and works in construction. He drinks two to three alcoholic beverages each week. He does not binge drink and has never had blackout episodes associated with drinking alcohol. He does not smoke, use electronic cigarettes, or other drugs. He has not been sexually active within the past year.

WHAT ADDITIONAL QUESTIONS WOULD HELP FORMULATE THE PROBLEM REPRESENTATION AND GENERATE AN INITIAL DIFFERENTIAL DIAGNOSIS?

Q1: Did the patient have upper respiratory or gastrointestinal symptoms?

Rationale for question: When a patient reports coughing up blood, it is critical to establish the source of the blood. The presence or absence of accompanying symptoms can help distinguish between hemoptysis — expectoration of blood from the lower respiratory tract¹ — and conditions that can mimic hemoptysis such as epistaxis, oropharyngeal bleeding, or hematemesis.² Such a distinction is especially relevant in this patient with a recent history of oral surgery.

Answer: The patient did not have epistaxis. Bleeding from the oral mucosa, including gingival bleeding and bleeding from the dental surgical site was not present. The patient was not experiencing odynophagia, dysphagia, nausea, or vomiting, nor hematemesis, abdominal pain, melena, or hematochezia.

Q2: How much blood has the patient expectorated?

Rationale for question: Understanding how much blood a patient has coughed up can help localize the source of

bleeding within the respiratory system and appropriately triage the acuity of the presentation. The lungs are highly vascularized, with two vascular systems that perform different functions. The low-pressure pulmonary circulatory system is primarily responsible for moving deoxygenated blood into the parenchyma where gas exchange can occur before returning to circulation through the pulmonary veins. Hemoptysis originating from the pulmonary circulation is typically scant and not life-threatening.³ It can be acute or chronic. In the latter situation, it is associated with infectious conditions such as tuberculosis (TB) and aspergillosis, inflammatory conditions such as interstitial pulmonary fibrosis, vascular conditions such as chronic thromboembolic pulmonary hypertension, and malignancies such as primary lung cancers or pulmonary metastases. In contrast, bleeding from the high-pressure bronchial circulation, which supplies blood to airway walls to the level of terminal bronchioles, the intima of large pulmonary arteries, intrapulmonary nerves, lymph nodes, and the visceral pleura, can cause large-volume bleeding into the airway that leads to respiratory compromise. It typically occurs in individuals with vascular anomalies such as aortobronchial fistulas and arteriovenous malformations. More than 200 ml of blood loss into the lungs over 24 hours is defined as massive hemoptysis, which is a life-threatening emergency that requires immediate airway evaluation to identify the source of hemorrhage and to stop the bleeding.³ Alongside quantitation, the clinical severity of the patient's presentation, such as the presence of airway compromise or hypoxemia, may also be used to triage patients to the appropriate level of care.⁴

Answer: The patient has only had one episode of hemoptysis, during which he expectorated about two spoonfuls of blood (<50 ml), and he has no evidence of airway compromise. This is considered small-volume, or nonmassive, hemoptysis.

Q3: Does the patient have a personal history of TB, or has he been in close contact with someone who might have TB?

Rationale for question: Active pulmonary TB is an important consideration in this case because it is a leading cause of hemoptysis worldwide and among people who have immigrated to the United States from regions of the world where TB is endemic.⁵ In the U.S. population, the incidence rate of TB is more than 13 times higher in people who were born abroad, such as this patient, compared with those born in the United States.⁶ It is therefore important to query the patient's risks for acquisition or reactivation of TB.

Answer: The patient does not have a personal history of TB and did not report any fevers, night sweats, or weight loss

that could be consistent with active TB. None of his family members or close contacts has ever been diagnosed with TB. He has never experienced homelessness, has not lived in a shelter or detention center, and has not had contact with the criminal justice system. Nonetheless, his history of emigration from a TB-endemic region puts him at high risk of prior exposure. Risk factors that may predispose patients to reactivation of previously acquired latent TB infection include chronic renal disease requiring hemodialysis, silicosis, diabetes, HIV, treatment with immunosuppressive therapy, and carcinoma of the head and neck.⁵ The patient had none of these risk factors.

PHYSICAL EXAMINATION

The patient's vital signs at the time of presentation are shown in [Box 2](#).

Box 2: Vital Signs

Temperature: 36.9°C (oral); heart rate: 72 beats per minute; blood pressure: 135/77 mm Hg; respiratory rate: 16 breaths per minute; and oxygen saturation: 99% breathing ambient air.

The patient appeared well and was breathing comfortably. He had no nasal deformities, and the nasal mucosa was dry with no evidence of recent bleeding. The oropharynx was moist with no ulcerations. His dentition was normal, and the site of recent dental surgery was clean. He had normal respiratory effort, and there were no adventitious sounds on auscultation. The heart rate was normal, with a regular rhythm and no murmurs. His chest pain was not reproducible on palpation. The abdomen was soft and nontender, and he did not have costovertebral angle tenderness. There were no rashes or palmar, plantar, or ungual lesions.

Part 2: Formulating the Problem Representation and Building the Differential Diagnosis

This is a young man with no known medical history other than recent oral surgery for an infected wisdom tooth. He had emigrated from Mexico a few years ago. He presents with pleuritic chest pain, cough productive of foul-smelling sputum, and nonmassive hemoptysis. An initial differential diagnosis can be focused on hemoptysis and would include infectious respiratory diseases, vascular abnormalities and malignancies,¹ as shown in [Box 3](#).

Box 3: Framework for Building the Differential Diagnosis for Hemoptysis

Infectious

Viruses: bronchitis

Bacteria: lung abscess (primary or secondary), necrotizing pneumonia, pulmonary nocardiosis, bronchiectasis exacerbation, septic emboli, odontogenic abscess

Mycobacteria: pulmonary TB, nontuberculous mycobacteria

Fungi: aspergillosis, mucormycosis, histoplasmosis, coccidioidomycosis

Parasites: paragonimiasis

Vascular

Cardiogenic pulmonary edema

Pulmonary embolism (PE)

Arteriovenous malformations

Aortobronchial fistula

Small-vessel vasculitides: anti-glomerular basement membrane (anti-GBM) disease, granulomatosis with polyangiitis (GPA), microscopic polyangiitis

Diffuse alveolar hemorrhage (DAH)

Malignancy

Bronchogenic carcinoma

Pulmonary metastases

Kaposi's sarcoma

Other

Foreign body aspiration

Potential infectious causes of hemoptysis include viral, bacterial, fungal, mycobacterial, and parasitic infections. A viral bronchitis is possible, although other common signs/symptoms, such as sore throat and wheezing, were absent in this patient. A lung abscess could have developed as sequela of the possible aspiration event during oral surgery, and would be consistent with foul-smelling sputum and pleurisy. Lung abscesses can be categorized according to duration of symptoms (acute/chronic), etiology (primary/

secondary), or mode of spread (bronchogenic or hematogenous). An abscess is considered chronic if symptoms persist longer than 6 weeks. Primary abscesses typically develop in healthy patients, often after an aspiration event, whereas secondary lung abscesses typically form in patients with underlying structural pulmonary pathology or immunocompromising conditions.

A third schema for classification is to distinguish the mode of spread, including bronchogenic abscesses formed as a result of aspiration and inhalation, or developing through hematogenous spread from a remote infected site such as dental infection, endocarditis, or septic thromboembolism. Necrotizing pneumonia is an uncommon sequela of community-acquired bacterial pneumonia and can cause hemoptysis as a result of the destruction of alveolar-capillary basement membranes. Having emigrated from a country in which TB is endemic, reactivation of latent TB is a primary consideration, although the lack of constitutional symptoms would be less consistent with active TB. The patient lives in the southwestern United States, putting him at risk for endemic mycoses, such as *Histoplasma* or *Coccidioides*. Environmental fungi, such as *Aspergillus* and *Mucor*, are ubiquitous in the soil and can be angioinvasive, causing especially severe disease in immunocompromised hosts. Because the patient has no known immunocompromising comorbidities or recent foreign

travel, the suspicion for nontuberculous mycobacterial or parasitic infections is lower; however, *Paragonimus mexicanus* is a parasitic species endemic to Mexico that can persist in humans for up to 20 years, causing cough, pulmonary structural changes, and hemoptysis.⁷

Vascular causes of hemoptysis can include direct structural vascular pathology, as occurs in aortobronchial fistulas, for example. In addition, small-vessel vasculitides, such as anti-glomerular basement membrane (anti-GBM) disease, granulomatosis with polyangiitis (GPA), and microscopic polyangiitis, can cause alveolar damage and subsequent hemorrhage; however, the lack of joint abnormalities or skin findings such as palpable purpura makes some of these diagnoses less likely. Malignancies such as bronchogenic carcinoma, pulmonary metastases, and Kaposi's sarcoma can cause hemoptysis through local angioinvasion. Although the patient had no clear increased risk factors for any malignancy, the possibility cannot yet be excluded.

WHAT ADDITIONAL INVESTIGATIONS WOULD BE HELPFUL?

A complete blood count with differential may show indices such as leukocytosis or leukopenia that could be consistent with systemic infection, or eosinophilia that could

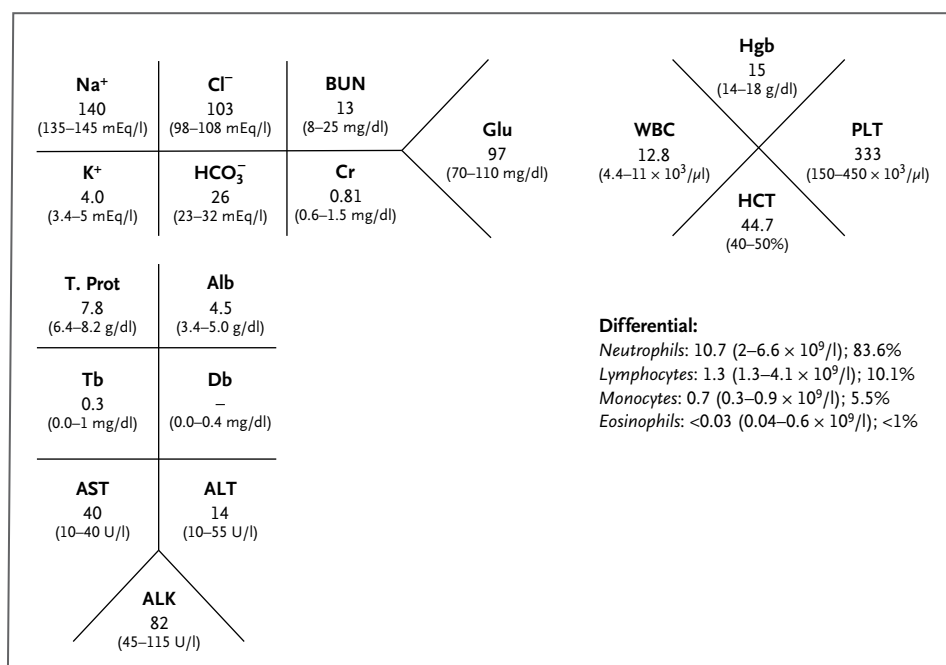


Figure 1. The Patient's Laboratory Results.

Reference ranges are given in parentheses. Alb denotes albumin; ALK, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; BUN, blood urea nitrogen; Cl⁻, chloride; Cr, creatinine; Db, direct bilirubin; Glu, glucose; HCO₃⁻, bicarbonate; HCT, hematocrit; Hgb, hemoglobin; K⁺, potassium; Na⁺, sodium; PLT, platelet; Tb, total bilirubin; T. Prot, total protein; and WBC, white blood cell.

be helpful in identifying a parasitic infection or coccidioidomycosis. Anemia may suggest chronic bleeding or an underlying inflammatory state. A comprehensive metabolic panel may identify signs of systemic inflammation affecting other organs (e.g., kidney, liver). Inflammatory markers such as C-reactive protein and erythrocyte sedimentation rate could be helpful in the evaluation for systemic infectious or inflammatory disease, such as TB, GPA, or anti-GBM disease, if clinical suspicion is high. A chest x-ray is a useful screening evaluation for anatomical abnormalities or lesions within the lung fields that may explain the presenting findings. An electrocardiogram is an essential component of the evaluation for chest pain and may provide corroborating information supporting a diagnosis of PE or congestive heart failure.

RESULTS

Results of the initial laboratory studies are shown in [Figure 1](#). Chest x-ray results are shown in [Figure 2](#). The electrocardiogram revealed normal findings with no evidence of ischemia.

WHAT ARE THE NEXT STEPS IN WORKUP?

Characterizing the cavitary lesion shown on the chest x-ray at higher imaging resolution may provide additional diagnostic input. Computed tomography imaging of the chest with intravenous contrast is appropriate for this purpose and was performed for the patient. Results are shown in [Figure 3](#).

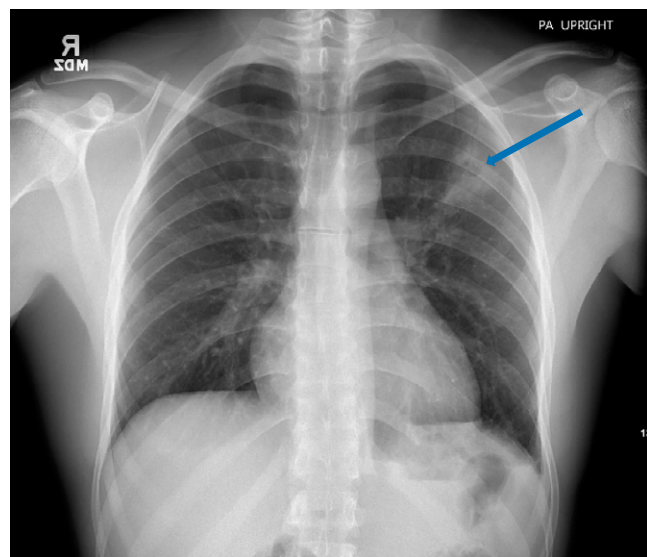


Figure 2. The Patient's Anteroposterior Chest X-Ray.

A thick-walled cavitary lesion is shown within the left upper lung (blue arrow).

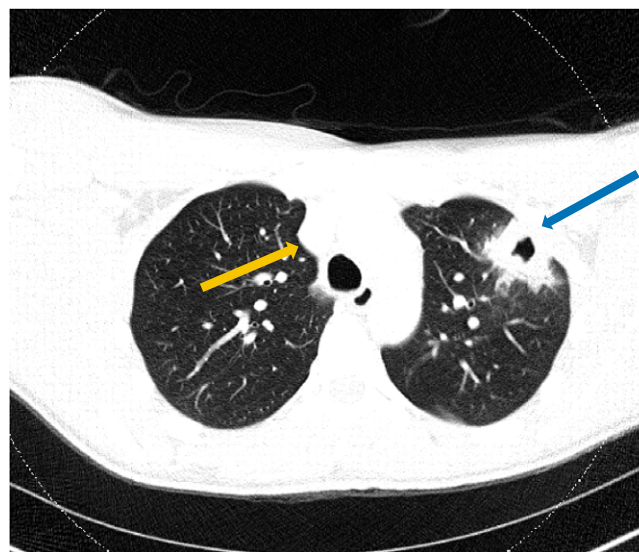


Figure 3. The Patient's Computed Tomography Imaging of the Chest with Intravenous Contrast.
A 3.7-cm thick-walled cavitary lesion (blue arrow) with surrounding ground glass opacities and prominent mediastinal lymph nodes (yellow arrow).

Part 3: Refining the Differential Diagnosis

The imaging findings help narrow the differential diagnosis to conditions that cause single, thick-walled pulmonary cavities. Two factors that help refine the differential diagnosis are the chronicity of the cavitary lesion and the patient's immune status. Acute and subacute cavities (<12 weeks) are often caused by acute infections (e.g., lung abscesses, necrotizing pneumonia, septic emboli in setting of right-sided infective endocarditis, fungal infections, nocardiosis), whereas chronic cavities (>12 weeks) are observed in chronic infections, malignancy, and small-vessel vasculitides associated with autoimmune conditions. Because this patient's clinical presentation was acute, there is no prior imaging available to establish the chronicity of the lesion. The patient's immune status is relevant because immunocompromised individuals are at higher risk of invasive infections with indolent bacteria and fungi. No structural changes consistent with bronchiectasis or other chronic pulmonary process were observed on imaging, which makes secondary abscess less likely. In addition, the lack of preceding pneumonia symptoms makes necrotizing pneumonia much less likely.

Pulmonary infarction after a PE can present with hemoptysis and a cavitary lesion, but the thick walls of the lesion make this less likely. No aortobronchial fistula or arteriovenous malformations were observed on computed tomography imaging of the chest. The single pulmonary lesion makes pulmonary metastases less likely. Primary bronchogenic carcinoma cannot be ruled out despite the absence of typical risk factors. HIV testing has not yet been obtained, and thus Kaposi's sarcoma remains on the differential. The refined differential diagnosis on the basis of these data is shown in [Box 4](#).

Box 4: Refining the Differential Diagnosis for Hemoptysis and a Cavitary Lung Lesion

Infectious

Viruses: ~~bronchitis~~

Bacteria: lung abscess (primary or secondary), ~~ne-~~
~~rotizing pneumonia~~, pulmonary nocardiosis, ~~bron-~~
~~chiectasis exacerbation~~, septic emboli, odontogenic abscess

Mycobacteria: pulmonary TB, nontuberculous mycobacteria

Fungi: aspergillosis, mucormycosis, histoplasmosis, coccidioidomycosis

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Vascular

Cardiogenic pulmonary edema

PE

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Small-vessel vasculitides: anti-GBM disease, GPA, microscopic polyangiitis

DAH

Malignancy

Bronchogenic carcinoma

Pulmonary metastases

Kaposi's sarcoma

Other

Foreign body aspiration

WHAT ADDITIONAL INVESTIGATIONS COULD BE HELPFUL?

Given the high clinical suspicion for an infectious process, investigations to identify potential microbial causes are an important next step. Blood cultures should be obtained to test for bacteremia associated with pulmonary infection. Gram stain and bacterial culture, acid-fast bacilli (AFB) stain and culture, and fungal wet prep and culture should be performed on sputum samples. Evaluation for TB acquisition, such as with an interferon- γ release assay (IGRA), should be performed.⁸ This patient likely received the Bacille Calmette-Guérin vaccine given the policy of vaccination for all citizens in his country of origin, Mexico.⁹ IGRA testing is preferred because it avoids potential false-positive results of TB skin testing in people who have received Bacille Calmette-Guérin vaccine.⁸ Of importance, a negative IGRA does not rule out the possibility of active TB, and specific testing for active pulmonary TB with AFB smear microscopy of three sputum samples, AFB sputum culture, and a nucleic acid amplification test on a respiratory specimen should all be obtained.¹⁰ Evaluation for fungal causes, such as *Aspergillus*, *Histoplasma*, and *Coccidioides*, should be performed. HIV testing should also be performed. Antinuclear antibodies, antineutrophil cytoplasmic antibodies, and anti-GBM antibodies may be useful in the assessment for small-vessel vasculitides. A transthoracic echocardiogram would be helpful to evaluate for signs of aortic pathology, left ventricular dysfunction, or vegetation suggesting indolent endocarditis.

SHOULD EMPIRICAL TREATMENTS BE STARTED?

A lung abscess is high on the differential diagnosis, and it would be reasonable to initiate an empirical antibiotic regimen to treat this infection pending additional workup as described in the following sections. Given the risk factor of recent dental infection and procedure, empirical treatment with ampicillin/sulbactam, which would treat infection from many oropharyngeal bacteria (including anaerobic organisms), would be appropriate. In cases in which TB remains on the differential diagnosis, it is prudent to avoid empirical fluoroquinolone treatment when possible to prevent selection of fluoroquinolone-resistant *Mycobacterium tuberculosis* strains.

HOSPITAL COURSE 1

The patient continued to have a cough that was productive of brown, foul-smelling sputum but had no additional episodes of hemoptysis. His vital signs remained normal, and the total white blood cell count normalized. Two samples of sputum

for Gram stain showed normal flora and moderate leukocytes. Routine bacterial culture did not grow any organisms. Airborne precautions were initiated, and AFB stains were performed on two samples of expectorated sputum. Results on the AFB stains were negative, *M. tuberculosis* polymerase chain reaction (PCR) in one sample was negative, and preliminary results on the AFB cultures were negative. The patient was unable to expectorate a third sputum sample for AFB stain/culture even after attempted sputum induction. Results of blood cultures obtained before initiation of empirical antibiotics remained negative at 5 days. Serologic work-up results revealed a nonreactive HIV test result, positive IGRA, and negative *Aspergillus* serum antigen, *Histoplasma* urine antigen, and *Coccidioides* serology testing. Antinuclear antibodies, antineutrophil cytoplasmic antibodies, and anti-GBM antibodies were negative. A transthoracic echocardiogram was performed, which revealed a 3-mm mobile echodensity in the mitral valve area that was interpreted to represent redundant chordal tissue (a normal variant).

HOW DOES THIS NEW INFORMATION AFFECT YOUR APPROACH TO THE CASE?

Although testing of sputum and blood samples has not revealed any organisms, infection from an indolent or difficult-to-culture organism remains possible. The presence of moderate or greater polymorphonuclear lymphocytes, even in the setting of normal flora, can be suggestive of pulmonary infection. A positive IGRA supports a diagnosis of latent TB with the possibility of reactivation. The results of fungal and autoimmune serology testing make endemic fungal infections and autoimmune conditions causing small-vessel vasculitis and DAH less likely. Negative results of bacterial blood cultures and transthoracic echocardiogram without evidence of right-sided infective endocarditis make septic emboli less likely.

HOSPITAL COURSE 2

As a result of the moderate clinical suspicion for pulmonary TB and the inability to obtain a third sputum culture, the patient underwent a bronchoscopy with bronchoalveolar lavage (BAL). BAL can be used for diagnosis in patients with suspected TB if sputum induction does not yield results.¹⁰ No endobronchial lesions or underlying structural abnormalities were seen during bronchoscopy. No organisms were identified on routine bacterial and fungal stains from BAL samples. Special efforts were made to recover anaerobic bacteria from BAL samples, but none was identified. AFB stains and *M. tuberculosis* PCR were negative from BAL fluid, and preliminary mycobacterial culture results

were negative. Cytologic analysis revealed erythrophagocytosis, hemosiderin-laden macrophages, and no malignant cells. At this point, the patient's cough had resolved, and he remained otherwise asymptomatic.

HOSPITAL COURSE 3

Given the negative initial test results for active TB and clinical improvement on ampicillin/sulbactam, the presumptive diagnosis of a primary lung abscess was made. The patient was discharged on treatment with amoxicillin/clavulanate to complete 6 weeks of antimicrobial therapy. At the time of follow-up in clinic, he was close to completing treatment and remained well without cough or additional hemoptysis. Results of all three AFB cultures remained negative. Repeat computed tomography imaging of the chest was planned after completion of the antibiotic course for posttreatment characterization of the cavitary lesion.

Part 4: Making the Diagnosis

FINAL DIAGNOSIS

The most likely diagnosis is hemoptysis from a primary lung abscess, possibly related to an odontogenic infection and aspiration event during removal of the infected tooth. The foul-smelling sputum, together with recovery after treatment with amoxicillin/clavulanate, is consistent with this diagnosis. Although no bacteria were shown on microbial testing, anaerobic bacteria and microaerophilic streptococci are the most common causes of lung abscess and can be difficult to culture. In addition, the patient's recent course of antibiotics taken after the dental procedure likely decreased the yield of the sputum culture. The positive IGRA, in the context of the patient's recent emigration from a TB-endemic region, and no microbiologic evidence of active TB — no AFB identified after sputum, culture, or PCR testing — is consistent with latent TB infection. The patient is scheduled for clinic follow-up to discuss the risks and benefits of receiving treatment for latent TB.

Take-Home Points

- Common causes of hemoptysis can be divided into infectious, vascular, and neoplastic. Early identification of the source of bleeding is critical for the management of massive hemoptysis.

- Lung abscesses may be categorized according to duration of symptoms (acute/chronic), etiology (primary/secondary), or mode of spread (bronchogenic or hematogenic). The most common microbial causes are mixed flora, including anaerobic and microaerophilic bacterial organisms.
- When patients from TB-endemic regions present for hemoptysis, it is prudent to consider a thorough evaluation for active TB, which includes querying symptoms such as fevers, night sweats, and weight loss and performing AFB smear microscopy and sputum culture on at least three sputum samples. Positive IGRA and negative AFB test results may suggest a latent TB infection, in which case the risks and benefits of treatment may be discussed with the patient in a shared decision-making approach. A negative IGRA does not rule out active TB infection.

Disclosures

Author disclosures are available at evidence.nejm.org.

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